

Data Mining Project : Mental Health

How do demographics and health traits predict mental health outcomes among Swiss medical students?

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Abstract

The study in this report examines mental health outcomes depression, anxiety, and burnout in a study sample of 886 Swiss medical students with the Carrard et al. (2022) dataset. The predictors were demographic factors (sex, age, year of study, mother tongue) and health characteristics (empathy, health satisfaction, use of therapy, motivation, hours of study). Five supervised classification models were evaluated using a 5-fold cross-validation, after a six-step preprocessing pipeline, which included cleaning, outlier capping, language grouping, feature engineering, and StandardScaler normalisation: Logistic Regression, Linear SVM, AdaBoost, XGBoost and Random Forest. The optimal test-set AUC of Logistic Regression in depression was 0.741, and in anxiety, the optimum test-set AUC of both Logistic Regression and Linear SVM were 0.760. Sub-scales on burnout provided AUC 0.672 (exhaustion), 0.675 (efficacy), and 0.590 (cynicism). The strongest predictors of all outcomes were therapy use and health satisfaction. In all the models, mother tongue was found to be not significant. Correlation with depression and anxiety is highly significant ($r = 0.73$), and Years 1-2 students represent the most vulnerable group of people.

Keywords

Mental health, medical students, depression, anxiety, burnout, machine learning, logistic regression, SVM, data mining, Switzerland

I. INTRODUCTION

Mental health deterioration among medical students is a well-documented global concern. The combination of intense academic workload, clinical responsibility, and personal development pressure makes medical education a uniquely high-risk environment. Swiss medical education spans six years across pre-clinical (Bmed 1–3) and clinical (Mmed 1–3) phases, placing particular stress on early-year students.

This project applies data mining and machine learning techniques to the Carrard et al. (2022) cross-sectional dataset to answer three research questions: RQ1- How do demographics and health traits relate to depression? RQ2 - How do they relate to anxiety? RQ3 - How do they relate to burnout across exhaustion, cynicism, and efficacy? The pipeline covers full EDA, outlier handling, feature engineering and systematic comparison of five classifiers across multiple feature configurations.

II. RELATED WORK

Research work on mental health among medical students is very common, especially when looking at factors like demographics and health. In this project, the related work is organised based on the three research questions (RQs). This structure helps to connect previous studies to the goals of this work.

For RQ1, many studies have examined depression among medical students using surveys and cross-sectional methods. For example, El Sayed S. A. et al. [2] used questionnaires to measure depression, stress, and anxiety among medical students in Egypt and found high levels of depression. Similarly, Al-Khani A. M. et al. [1] studied sleep quality and found that poor sleep is linked to higher levels of depression. In Switzerland, Carrard V. et al. [10] found that students who are less satisfied with their studies tend to have worse mental health. These studies mostly use surveys and simple statistical methods like correlation and regression. They show that factors like sleep, stress, and academic pressure are related to depression. However, the main limitation is that they use cross-sectional data, so they cannot prove cause and effect. This study builds on past research by looking at many factors together to better understand depression.

For RQ2, research on anxiety is often studied together with depression, but it also has its own patterns. El Sayed S. A. et al. [2] also found high levels of anxiety among students. In another study, Galán J. et al. [4] used statistical analysis to study anxiety, burnout, and empathy, and found that anxiety is strongly linked to emotional stress and burnout. Similarly, Kaur P. et al. [5] showed that empathy can affect anxiety in both positive and negative ways. These studies use similar methods and show that anxiety is affected by both personal traits and external stress. However, many of them do not study anxiety on its own, which makes it harder to understand its specific causes. This study improves on that by looking at anxiety as a separate outcome.

For RQ3, burnout is another key issue in medical education. Williams J. et al. [3] studied burnout, anxiety, and depression and found that burnout is strongly linked to poor mental health. Afolabi O. A. et al. [7] also found that lifestyle and stress are important factors in burnout. In a broader study, Dyrbye L. N. et al. [8] showed that workload, sleep, and well-being are key factors affecting burnout. Other studies show that burnout can differ across countries. For example, Moreira D. de A. and Furtado E. de S. [6] found differences between US and Brazilian students. Also, Matthews L. and

Quin J. [9] used interviews to show how students experience stress and burnout in daily life. Most studies treat burnout as one single concept. A limitation is that they do not always look at its parts, like exhaustion, cynicism, and efficacy. This study improves on that by looking at burnout in more detail.

Past research shows that depression, anxiety, and burnout are affected by many factors such as sleep, stress, and personal traits. Most studies use similar methods and often look at only a few factors at a time. This study adds to existing research by combining demographic and health traits and focusing on Swiss medical students, giving a more complete understanding of mental health outcomes.

III. OBJECTIVES

The objective of this study is to explore mental health patterns among medical students in Switzerland. It focuses on three key outcomes: depression, anxiety, and burnout using survey data. The study also aims to identify demographic patterns and possible risk groups by examining factors like sex, age, year of study and mother tongue.

To achieve this, the study takes a three-part approach based on these outcomes: depression, anxiety, and burnout. Each part is guided by a specific research question (RQ) and hypothesis (H).

The first objective is to examine how demographic and health traits relate to depression among medical students. Based on this objective and the literature reviewed, the following research question and hypothesis are defined:

RQ1: *How do demographics and health traits relate to depression among medical students in Switzerland?*

H1: Mental health outcomes differ across sex, meaning that male, female, and non-binary students may show different levels of depression.

The second objective is to analyse how demographic characteristics and health traits relate to anxiety. From this, the following research question and hypothesis are formed:

RQ2: *How do demographic characteristics and health traits relate to anxiety among medical students in Switzerland?*

H2: Age is related to mental health, meaning that younger students may report higher levels of anxiety.

The third objective is to explore burnout and its main dimensions, including exhaustion, cynicism, and efficacy, in relation to demographic and health traits.

RQ3: *How do demographic characteristics and health traits relate to burnout (exhaustion, cynicism, and efficacy) among medical students in Switzerland?*

H3: Students in earlier years of study may report higher levels of burnout symptoms.

H4: Students with different mother tongues may show

differences in burnout levels, although no strong effect is expected.

The research questions focus on relationships between variables rather than complex predictions. Based on previous studies, it is expected that female students may report higher levels of depression, anxiety, and psychological distress than male students. Younger students and those in earlier years of study are also expected to show higher levels of mental health symptoms. In addition, there may be some differences across mother tongue groups, although no strong effect is expected.

However, the goal is not only to test these expectations, but also to explore patterns in the data and identify possible risk groups. By analysing these relationships, the study aims to provide deeper insights into mental health among Swiss medical students.

IV. DATA

This study uses the Medical Student Mental Health Dataset from Kaggle, created for research and educational purposes. The dataset focuses on medical students in Switzerland and aims to examine mental health, empathy, and burnout using survey-based and psychological assessment data. It is designed to reflect real-world patterns in student well-being.

The dataset contains multiple records and 18 variables in CSV format, covering demographic information, lifestyle factors, psychological indicators, and mental health outcomes. Variables include age, year of study, sex, mother tongue language, study hours, health status, and several psychological scale scores such as depression (CES-D), anxiety (STAI), and burnout (MBI).

Table 1 shows a detailed description of the dataset variables.

Each row in the dataset represents one participant, while each column represents a specific variable. The dataset supports the study's objective of identifying patterns in depression, anxiety, and burnout.

TABLE I. Dataset Variables

S/N	Variable	Type	Description
1	age	Integer	Age of participant
2	year	Integer	Year of study
3	Sex	Categorical	Gender of the participant
4	glang	Categorical	Mother tongue/language spoken
5	job	Categorical	Job status of the participant
6	Stud_h	Integer	Study hours per week
7	health	Categorical	Self-reported health status
8	psyt	Integer	Psychological distress score
9	cesd	Integer	Depression score

10	stai_t	Integer	Anxiety score
11	mbi_ex	Integer	Burnout – Exhaustion
12	mbi_cy	Integer	Burnout – Cynicism
13	mbi_ea	Integer	Burnout – Professional efficacy
14	qcae_cog	Integer	Cognitive empathy score
15	qcae_aff	Integer	Affective empathy score
16	erec_mean	Integer	Empathy rating mean score
17	amsp	Integer	Academic motivation score
18	jspe	Integer	Job satisfaction score

A. EXPLORATORY DATA ANALYSIS

Variables were classified into categorical and numeric types to support analysis. Distribution plots for both variable types are shown in Figure 1 and Figure 2. Bar charts were used to examine distributions, while box plots were used to analyse key mental health outcomes (depression, anxiety, and burnout) and identify possible outliers as shown in Figure 2.

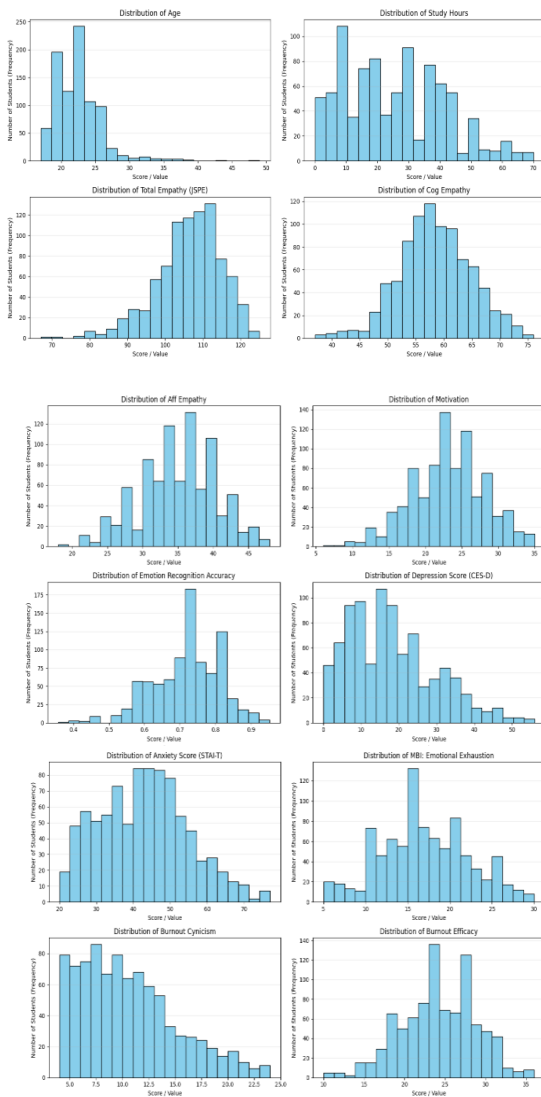


Fig. 1. Distribution of numerical variables

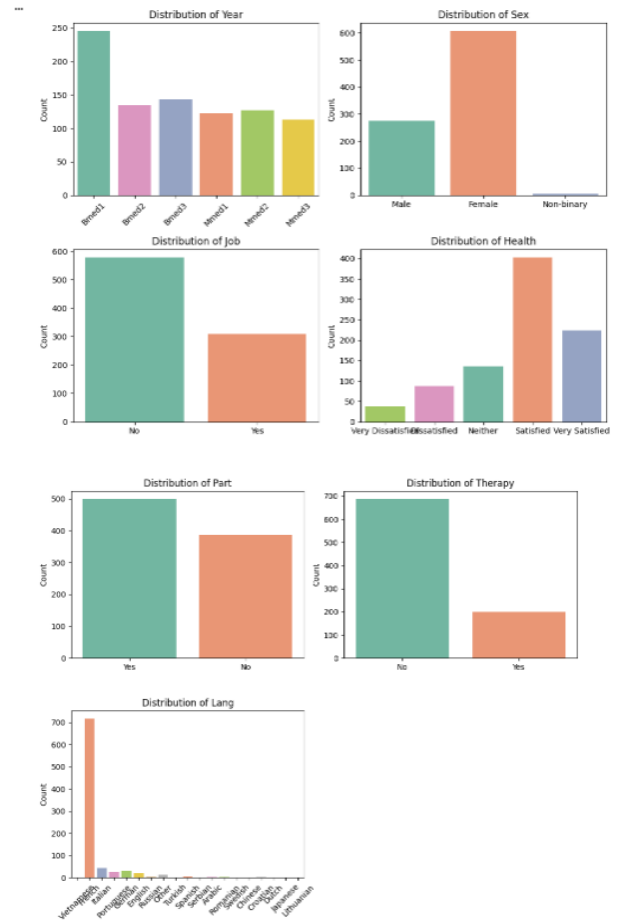
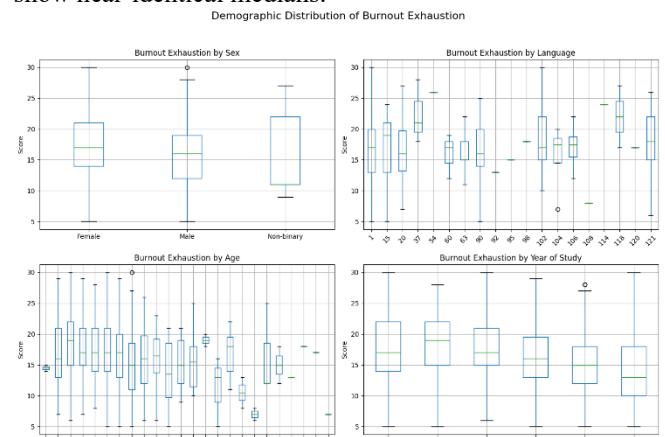


Fig. 2. Distribution of categorical variables

Demographic group analysis: Box plots compared depression, anxiety and burnout exhaustion across sex, age, year of study, and language. All three outcomes peak in Years 1-2 and decline through Year 6. Females show slightly higher median, males show more extreme outliers. Language groups show near-identical medians.



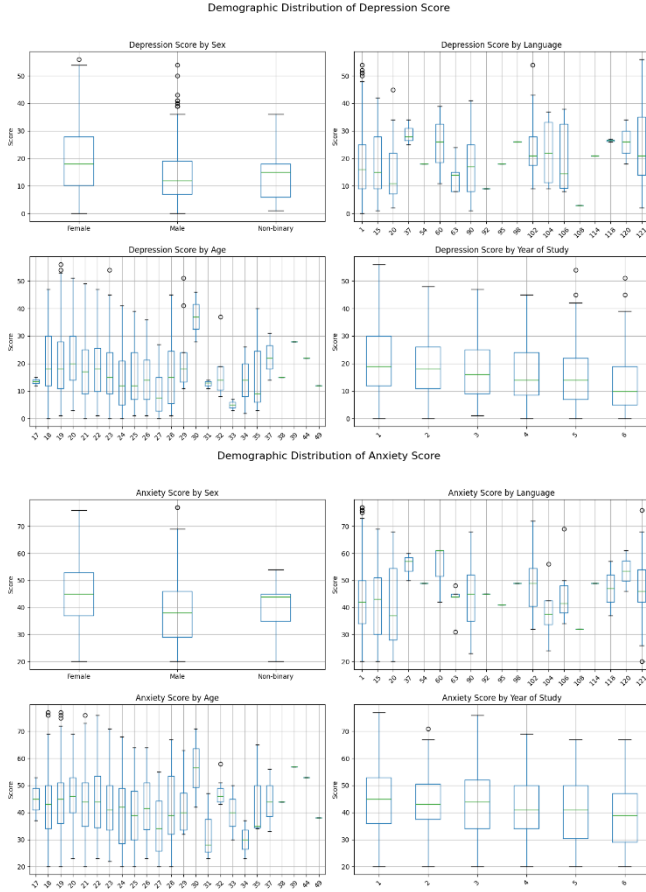


Fig. 3. Boxplot by Demographic Groups

B. PRE-PROCESSING

The preprocessing pipeline transformed raw survey data into a high-quality modeling input by implementing feature engineering for age and study hours, followed by Z-score standardization. To ensure model robustness, we applied 1% Winsorization to manage outliers and utilized SMOTE to rectify the significant class imbalance between healthy and distressed students as summarized in the table II below.

TABLE II. Pre-Processing Pipeline - All Steps

Step	Action	Method	Key Outcome
1	Cleaning and renaming	Rename, remove 3 non-binary	883 records, clear column names
2	Variable classification	Manual mapping dicts	Categorical vs. numerical separated
3	Outlier detection	IQR box plots, 18 variables	Age: 24 outliers; empathy: 9; motivation: 2
4	Outlier mitigation	Age cap=35; Winsorize 1%/99%	7 students (2.71%); features within +/-3 sigma
5	Language grouping	40+ codes to 4 Swiss groups	French 717, Other 93, Italian 45, German 31
6	Feature engineering	Age binary, study hrs groups; one-hot	20 final features, binary targets created
7	Split and scaling	75/25 split, StandardScaler (train only)	Train=660, Test=221; no data leakage

Outlier detection : IQR box plots across all 12 numerical variables. Data integrity audit confirmed mature students aged 36-49 are valid participants - year of study and psychological scores fall within normal ranges. Age was capped at 35 (not removed) to preserve all 886 records, affecting only 7 students (2.71%). Mental health target scores kept raw to preserve clinical signal.

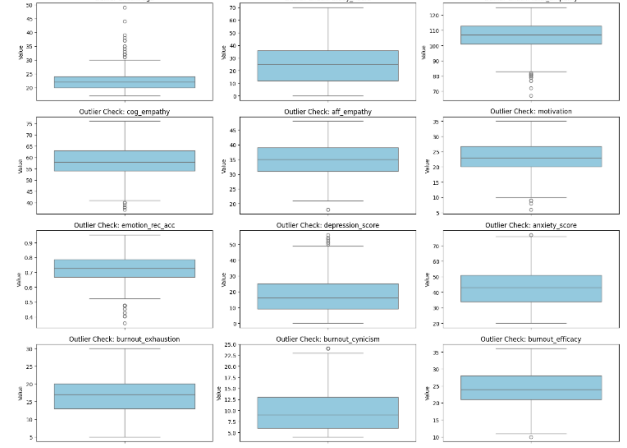


Fig. 4. IQR Outlier Detection - Box Plots Across All Numerical Variables.

Language grouping: 40+ codes consolidated into 4 Swiss-contextual groups. Box plot validation confirmed near-identical median depression scores across all groups (approximately 15-20), confirming no artificial bias is introduced by the grouping.

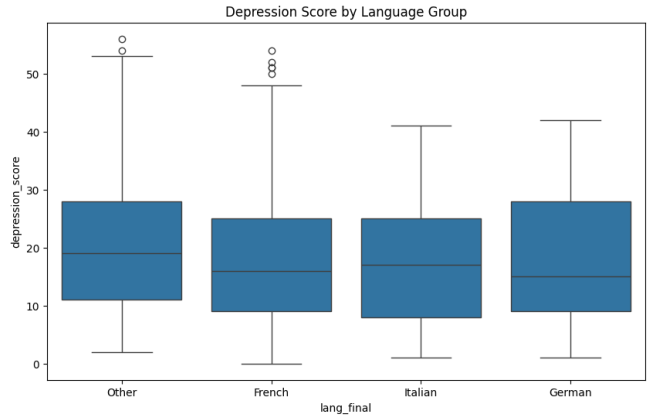


Fig. 5. Language Grouping Validation-Depression Score by Language Group

Correlation analysis: Pearson correlation matrix confirmed very strong co-movement between depression and anxiety ($r = 0.73$), strong depression-exhaustion link ($r = 0.63$), and a protective inverse relationship between efficacy and depression ($r = -0.45$). The scatter plot of anxiety vs. depression (Fig. 6) confirms a clear linear trend.

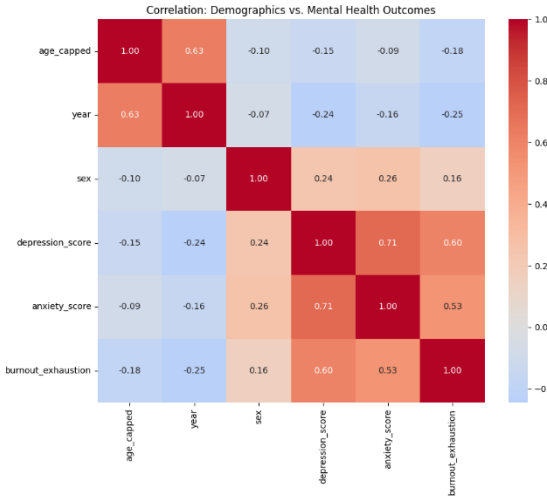


Fig. 6. Correlation Heatmap

V. METHODS

The research questions focus on predicting three mental health outcomes: depression, anxiety, and burnout scores, which are all numeric. Initially, the task was designed as regression, but in real-world scenarios, the health data is often noisy, and predicting a precise number is quite difficult. Therefore, we turned the problem into classification by putting thresholds to the target variables.

Data and Features

After splitting the dataset into training and test sets using a 75/25 ratio, the features were standardized using a Standard Scaler. The scaler was fit only on the training set and then applied to the test set to avoid data leakage. The input features include year, sex, lang_final_French, lang_final_German, lang_final_Italian, lang_final_Other, part, job, study_hours, health, therapy, total_empathy, cog_empathy, aff_empathy, motivation, emotion_rec_acc.

Models and Experiments

Logistic Regression, SVM, Random Forest, AdaBoost, and XGBoost were trained for each target variable. To handle class imbalance, all classifiers were initialized with the parameter `class_weight='balanced'`. Moreover, we performed various experiments on all these models such as training on all features, applying SMOTE, removing language features, converting numerical variables (age and study hours) into categories, and training on selected top features.

For the best-performing models, hyperparameter tuning was done using GridSearchCV with 5-fold cross-validation, tuning the regularization parameter C over [0.001, 0.01, 0.1, 1, 10, 100].

One feature engineering technique that we performed involved categorizing age and study hours. Age was split using a cutoff of 23, and study hours were divided into three categories: low, moderate, and high. We also extracted the feature importance after training using coefficients for

Logistic Regression. Later, the top features were used in another experiment to see if a smaller feature set could enhance the performance.

Target Modeling : Depression and anxiety were treated as binary classification tasks. Depression was turned into a binary target using the cutoff 16 for men, and 21 for women. We also used 44 as the threshold for anxiety.

Burnout was modeled through its three scores: exhaustion, cynicism, and efficacy. Each label was treated as a separate binary classification task. For categorizing the above labels, we applied the official Maslach cutoffs, but since they showed an extreme class imbalance. We then switched to data-driven cutoffs based on mean and standard deviation computed from the training set. These are the cutoffs for the burnout targets: exhaustion (≥ 19.44), cynicism (≥ 12.21), and efficacy (≤ 26.74 indicating high burnout).

Since burnout consisted of three dimensions, we followed two paths: classifier chain and multi-output classification. In the classifier chain approach, the predictions of one dimension are used as additional inputs for the next, whereas in multi-output classification the labels are independent, and separate models are trained for each dimension.

Evaluation

All models were trained and evaluated using 5-fold cross-validation during training. Later, the best model was chosen to be evaluated on the test set. The main metric to compare the model performances was AUC (Area Under the Curve), since it is suitable for classification with class imbalance.

VI. RESULTS

As mentioned in the previous section, five models were evaluated on each target: Logistic Regression, Linear SVM, Random Forest, AdaBoost and XGBoost. Cross-validation with 5 folds was also applied on the training data.

A. Depression

Given the depression cutoff, the distribution of depression across the whole dataset was balanced, with 41% of students at risk of depression and 59% not. Across all experiments, Logistic Regression and Linear SVM consistently showed the highest AUC scores, while Random Forest performed the weakest. SMOTE, removing language features, and categorizing age and study hours all produced only minimal changes in performance. The best result was achieved by Logistic Regression trained on the top features, reaching a CV AUC of 0.727 as shown in Table III. This model was then evaluated on the test set with 0.7416 as the AUC. The feature importance on Logistic Regression was also extracted as shown in Fig. 7.

TABLE III. CV AUC scores for Depression. All = all features, No Lang = language features removed, No Lang + Age = language and categorized age

removed, No Lang + Age/Hrs = language, age, and study hours categorized, Top = top selected features.

Model	Experiment					
	All	SMOT E	No lang	No lang + age	No lang + age/hr s	Top
Logistic regression	0.7186	0.7122	0.7189	0.7221	0.7222	0.7272
Linear SVM	0.7183	0.7100	0.7075	0.7096	0.7109	0.7154
Random Forest	0.6870	0.6884	0.6779	0.6852	0.6951	0.6603
AdaBoost	0.7130	0.7099	0.7130	0.7156	0.7175	0.7195
XGBoost	0.7009	0.6974	0.6976	0.7020	0.6994	0.6994

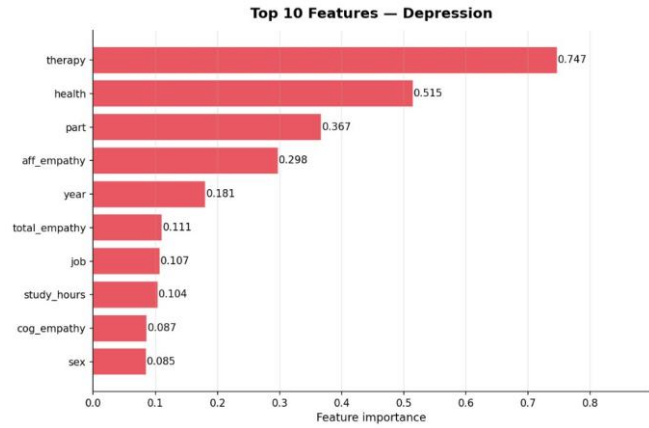


Fig. 7. Feature importance for depression

B. Anxiety

The anxiety target distribution was also balanced with 48% of students showing anxiety and 48% of students not. Anxiety was generally predicted better than depression across all models and experiments. Linear models outperformed tree-based ones consistently. The best result was Logistic Regression on the top features with a CV AUC of 0.768. Similar to depression, feature engineering steps had only a small effect, and SMOTE did not bring consistent improvement (Table IV). The best model was evaluated on the test set with 0.8121 as AUC. The feature importance is illustrated in Fig. 8.

TABLE IV. CV AUC scores for Anxiety.

Model	Experiment					
	All	SMOT E	No lang	No lang + age	No lang + age/hr s	Top
Logistic regression	0.7604	0.7557	0.7602	0.7599	0.7555	0.7668
Linear SVM	0.7601	0.7563	0.7590	0.7588	0.7544	0.7655
Random Forest	0.7086	0.7197	0.7197	0.7252	0.7226	0.7354
AdaBoost	0.7493	0.7401	0.7461	0.7486	0.7417	0.7563

XGBoost	0.7402	0.7416	0.7424	0.7429	0.7446	0.7450
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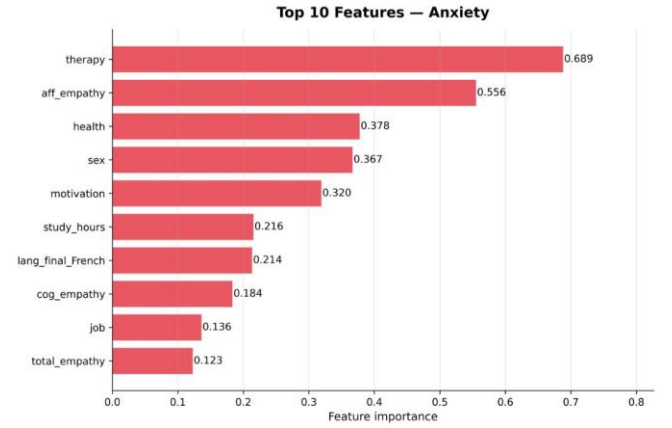


Fig. 8. Feature importance for anxiety

C. Burnout

Using data-driven binary cutoffs, exhaustion and cynicism were relatively imbalanced, with 30% and 26.4% of students classified as burned out respectively. Efficacy was the opposite, with 65.8% of students showing high burnout.

Burnout was the most challenging target overall as it consists of three dimensions. Therefore, it was modeled in two ways: multi-output classification and classifier chain.

Multi-output classification: In this approach, Logistic Regression performed best across experiments. Exhaustion was the most predictable label, while cynicism and efficacy were harder to classify. Removing language features and categorizing age or study hours had minimal impact on performance. The results are illustrated in Table V.

Classifier chain: The same experimental setup was repeated in this method. The classifier chain did not clearly outperform multi-output classification. Logistic Regression remained the best model, and the ordering exhaustion > cynicism > efficacy did not bring a noticeable improvement compared to the previous approach.

The best burnout model was evaluated on the test set per dimension, with exhaustion achieving a test AUC of 0.708, cynicism 0.612, and efficacy 0.636. Overall, burnout was harder to predict compared to depression and anxiety, which may reflect the more complex and multi-dimensional nature of the outcome. The feature importance along all three dimensions is shown in Fig. 9.

TABLE V. Multi-output classification - macro AUC scores for Burnout.

Model	Experiment			
	All	No lang	No lang + age	No lang + age/hr s
Logistic regression	0.6557	0.6557	0.6556	0.6491
Linear SVM	0.6119	0.6119	0.6124	0.6137
Random Forest	0.6096	0.6096	0.6155	0.6125
AdaBoost	0.6389	0.6389	0.6415	0.6312

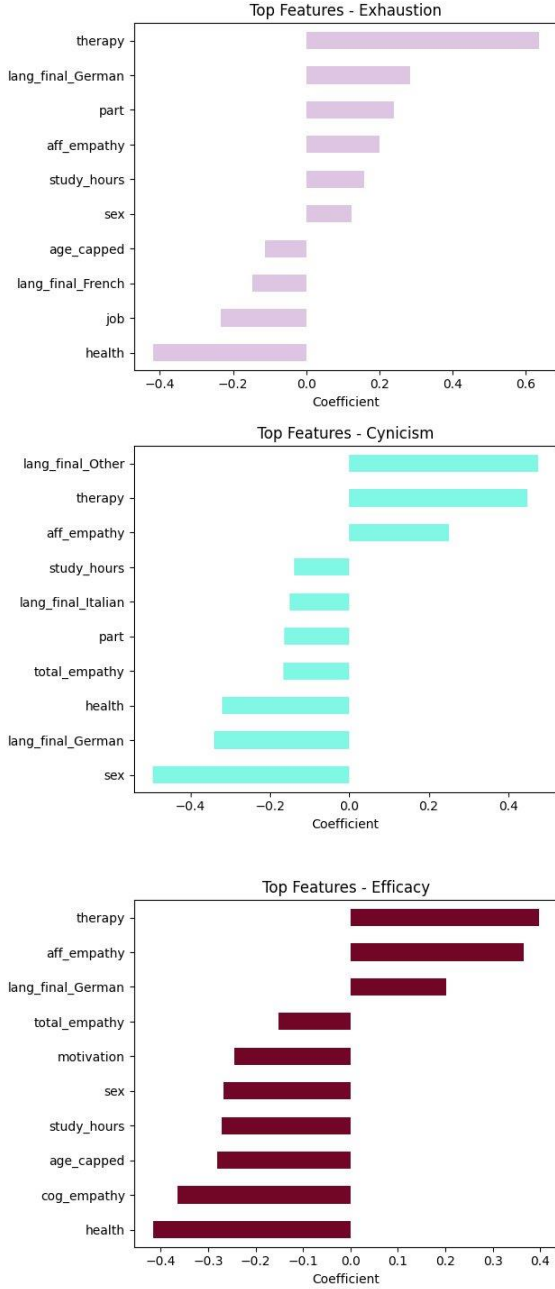


Fig. 9. Feature importance across burnout dimensions

VII. DISCUSSION

The results indicate that Linear Regression and SVM outperformed tree-based models across all targets. This suggests that the relationship between the input features and mental health outcomes is mostly linear. Language is not an informative predictor as removing it from the dataset had only a very subtle effect on performance. This is worth noting because the dataset is dominated by French-speaking students, meaning the language feature carries very little variance and therefore very little signal.

For depression, based on the feature importance results, therapy (coefficient 0.758), health (0.514), and partner status

(0.375) were the top three features in Logistic Regression. This is meaningful because whether a student is already in therapy can be seen as a clue for existing mental health struggles, health shows overall wellbeing, and having a partner may relate to social support. Affective empathy also appeared across models, which suggests that the way students emotionally connect with others plays a role in their mental state.

For anxiety, therapy (0.686), affective empathy (0.553), and health (0.384) were the top three features. This can suggest that people who are more emotionally reactive and have higher empathy may experience higher anxiety levels.

For exhaustion, therapy had the strongest positive coefficient, followed by German language, while health, and job were negative. The positive effect of German language may reflect differences between language groups in the Swiss medical system, however, this might not be very trustworthy since the languages feature is highly imbalanced.

For cynicism, other languages and therapy were the strongest positive predictors while sex and German language were negative.

For efficacy, therapy and affective empathy were the top positive predictors, while health and cognitive empathy were negative. The opposite signs of affective and cognitive empathy are interesting.

One clear pattern across all targets is that therapy, health, and affective empathy are the most consistently important features. This indicates that emotional wellbeing and health conditions are the strongest predictors for mental health risk among these students. Furthermore, demographic features like age and year of study had relatively low importance across all models.

Ultimately, our findings are somehow consistent with the related work. Therapy, health and empathy were the most predictive features, and based on El Sayed S. A. et al. [2] and Carrard V. et al. [10] health-related and personal factors are strongly related to depression and anxiety in medical students. Moreover, the role of empathy aligns with Kaur P. et al. [5], who showed that empathy can both protect against and contribute to mental health difficulties.

Limitations

The dataset used in this study is quite small and is mostly made up of female students whose mother tongue is French. Because of this imbalance, the impact of cultural background and gender differences cannot be properly studied and analyzed. Moreover, health datasets are often noisy and self-reported scores such as health in this dataset can be quite

arbitrary as different people may have different opinions about health scales. The binary cutoffs used to categorize the targets are also somewhat arbitrary and may not generalize well to other populations and settings.

Future work

Collecting more data from international students and other genders would be a great step forward, as different cultures and genders can show different mental health outcomes. Additional features such as academic performance (e.g. grades) and sleep quality would also be interesting to add.

VIII. REFLECTION ON GROUP WORK

Fatima Olasunkanmi-Ojo: Data understanding and column renaming, including the classification of variables into categorical and numeric types to support correct analysis. Distribution analysis was performed using bar charts, while box plot analysis was used for key mental health outcomes (depression, anxiety, and burnout). These visuals formed the foundation of the exploratory data analysis (EDA).

Shalini Singh: IQR outlier detection across all 18 variables, and carried out data integrity checks, including auditing and validating mature-student outliers, language consolidation from 40+ codes to 4 groups, validation via depression box plots, and production of the correlation matrix and scatter plot. Additionally, managed administrative tasks, including coordinating team activities and maintaining clear communication among group members.

Mahya Ehsanimehr: ML implementation, one-hot encoding, 75/25 split, StandardScaler pipeline, all five classifiers for RQ1-3, grid search, SMOTE experiments, multi-output classifier, and classifier chain for burnout.

The most interesting finding was that therapy use and health satisfaction appeared as top two predictors across every RQ. The most challenging aspect was burnout cynicism, which refused to yield a meaningful AUC despite every modelling approach, highlighting the limits of demographic prediction for emotional detachment.

Task allocation was based on individual strengths and interests. Responsibilities were clearly divided across data preprocessing, exploratory analysis, and machine learning implementation, while maintaining regular communication to ensure alignment and integration of work. Each group member contributed significantly within their area of responsibility and the collaborative effort was key to completing the project successfully.

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